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Dynamic event-triggered optimized control for nonlinear multi-agent systems via reinforcement learning

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A B S T R A C T

The practical deployment of high-order nonlinear multi-agent systems (MASs) is often hindered by two critical bottlenecks: limited communication resources and unavoidable environmental uncertainties. To overcome these challenges, an innovative optimized consensus tracking control framework with a dynamic event-triggered mechanism is established. For handling the unknown nonlinearities, multilayer perceptrons (MLPs) are employed as adaptive function approximators, and their parameter tuning is guided by a multi-agent Actor-Critic reinforcement-learning (RL) mechanism embedded with consensus information. The optimized controller is equipped with a dynamic event-triggered control (DETC) mechanism that adaptively regulates sampling-error thresholds online, thereby reducing the burden on data exchange and processing. By utilizing the Lyapunov analysis method, closed-loop stability is ensured and Zeno behavior is effectively prevented. To demonstrate the performance of the presented approach, numerical studies are conducted on a representative multi-electromechanical system. The results show that the proposed method not only significantly reduces communication overhead but also demonstrates superior robustness against sensor noise compared to existing independent learning approaches, providing a reliable solution for resource-constrained networked control systems.

1. Introduction

Over the past few decades, control strategy design and analysis have gained extensive attention, with applications extending to fields like spacecraft formation flying (Zhang et al., 2024a), unmanned aerial vehicles (Rezaee & Abdollahi, 2014; Tian et al., 2025) and smart grids (Dong et al., 2024). This extensive attention has driven researchers to conduct a great deal of in-depth studies on various control schemes for MASs. Consensus control, as a crucial research direction in MASs, is a strategy that enables a group of agents to eventually synchronize their states or outputs to common trajectories, or to satisfy a preset consensus relationship. However, the majority of existing consensus control strategies are developed under the ideal assumption of linear agent dynamics, such as Li et al. (2015) and Trentelman et al. (2013). In practical applications, MASs are often inherently nonlinear, and may contain uncertainties and unknown dynamics. These factors significantly complicate the analysis and design of control strategies, making the consensus problem for nonlinear MASs a more challenging yet practically important field of research.

The control strategy proposed above requires continuous communication, which can effectively ensure stability and performance, but imposes significant burdens on network bandwidth and computing resources. To alleviate this limitation, event-triggered control has emerged as a promising alternative solution (Liu et al., 2021). Its core idea is to save communication and computational resources by executing control tasks only when a system state-dependent threshold condition is violated. To further enhance this efficiency, recent research has introduced DETC, where the triggering threshold is dynamically adjusted according to the system states, leading to fewer triggers while maintaining desired performance (Yang et al., 2025; Zhang et al., 2024b). As a core area of modern control science, optimal control significantly influences intelligent control research and is extensively used in industrial engineering (Guo et al., 2020; Wu et al., 2019). Achieving optimal control typically involves solving the HJB equation. However, the nonlinear and complex form of the HJB equation poses substantial challenges for obtaining its solution. To address this issue, researchers have extensively and successfully applied RL to solve optimal control problems (Huang et al., 2026). RL algorithms usually adopt the Actor-Critic neural networks

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architecture. Under this framework, the Actor network interacts with the environment, while the Critic estimates the value function and guides the Actor's policy updates.

Although RL methods have effectively solved the optimal control problem, accurately modeling the unknown nonlinear terms in the system remains a major challenge. To address this challenges, this paper proposes a novel distributed control framework. We integrate a DETC with RL based control law. Specifically, we utilize Multilayer Perceptrons (MLPs) within an Actor-Critic architecture to handle the high-order nonlinear dynamics and unknown system parameters.

The primary contributions of this study are outlined below:

1. A learning-aware DETC is proposed. Compared with existing schemes (Zhang et al., 2022a,b), the proposed DETC threshold is deeply co-designed with the adaptive neural networks. Specifically, the dynamic threshold integrates not only the consensus tracking errors but also the evolving neural weights. This "learning-aware" logic allows the threshold to dynamically relax as the Actor-Critic evaluation stabilizes, thereby optimally balancing cooperative control performance and communication resource conservation in real time.
2. This paper represents an innovative application of consensus RL to high-order MASs. Unlike the optimal control schemes in Shi et al. (2022), Wen and Chen (2023) and the strategy proposed in Li et al. (2021), this paper adds neighbor interaction information to the Actor-Critic network design, which allows for stable estimation of key parameters and ultimately achieves global parameter consensus. As a result, agents can realize global consensus through local communication.
3. In contrast to the RBF neural networks employed in Xu et al. (2023), this paper introduces the use of MLPs as a more powerful and adaptive solution for systems control. The key innovation lies in their elimination of the need to manually redesign basis functions. Instead, MLPs dynamically adapt to system changes through real-time weight updates, effectively addressing the dynamic and complex challenges in MASs that have hindered fixed-structure networks like RBF.

2. Related work

2.1. Event-triggered mechanisms

In networked MASs, continuous communication between agents and controllers is often impractical due to limited bandwidth and energy resources (Salehi et al., 2025). To mitigate this, event-triggered control (ETC) has emerged as an efficient alternative to traditional periodic sampling (Ruan et al., 2025b). Under the ETC framework, data transmission is only executed when a predefined triggering condition is violated, thereby significantly reducing the communication load (Zhao et al., 2023). Recently, the research has evolved from static event-triggered mechanisms to DETC. As highlighted in Ruan et al. (2025a), DETC introduces a dynamic variable into the threshold, which allows for a more flexible and less frequent triggering condition compared to static schemes. For example, the authors in Xu et al. (2023) explored dynamic triggering for partially unknown nonlinear systems to optimize resource utilization. Nevertheless, for high-order systems with multiple layers of uncertainty, designing a DETC that balances control precision with communication efficiency remains a non-trivial task. This paper addresses this gap by proposing a performance-aware DETC mechanism where the triggering threshold is adaptively regulated by the system's learning states and tracking errors.

2.2. Reinforcement learning-based optimal control

Optimal consensus control for nonlinear MASs remains a significant challenge (Dong et al., 2021), as it necessitates solving the complex HJB equation to obtain the optimal control law. To address this computational hurdle, Shen et al. (2024) proposed online Q-learning algorithms to solve the HJB equation. In Zhu et al. (2025), RL-based backstepping controllers were designed to tackle the challenge of optimal control law design in nonlinear MASs. In Zhu et al. (2023), backstepping was incorporated with an adaptive RL algorithm to address stochastic nonlinear systems, where an identifier critic actor structure enabled optimal control. Although previous papers have successfully designed optimal control laws, they only considered local information and ignored global information. This constraint leads to agents making decisions based on their own limited observations, consequently compromising the overall system performance and preventing it from reaching a globally optimal level.

2.3. Neural network-based function approximation

Accurately modeling the unknown nonlinear terms in the system remains a major challenge in the control of MASs. The core difficulty lies in describing such nonlinear dynamics using precise mathematical expressions. To address this, data-driven techniques, including neural networks and fuzzy logic systems, have been extensively employed to approximate uncertain dynamics and predict complex behaviors across various domains (Shen et al., 2025a,b). For instance, Salehi and Babaei (2025) utilized network-based analysis for optimizing national management strategies, while Salehi and Khedmati (2024) demonstrated the effectiveness of integrated predictive models and fuzzy clustering in identifying at-risk patients. Similarly, in the field of financial forecasting, Jin and Xu successfully applied neural networks to predict commodity price indices (Jin & Xu, 2025a) and high-frequency futures prices (Jin & Xu, 2025b), showcasing the powerful approximation and generalization capabilities of these methods in handling unknown nonlinear patterns.

In the context of control systems, studies have adopted RBF networks to approximate unknown nonlinear terms. However, the performance of RBF networks depends strongly on the selection of kernel parameters, which limits their applicability in highly dynamic environments. Other approaches have employed fuzzy logic systems to model uncertain agents. Nonetheless, fuzzy systems often rely heavily on manually designed rules and parameters, restricting their ability to autonomously learn high-dimensional patterns directly from data. To overcome these limitations, this paper introduces the use of MLPs. Unlike fixed-structure networks, MLPs eliminate the need to manually redesign basis functions and can dynamically adapt to system changes through real-time weight updates, thereby providing a more adaptive solution for high-order nonlinear MASs.

This study is structured as follows. Section 3 formulates the preliminary knowledge and problem formulation. Section 4 develops the backstepping based optimal controller, formulates the event-triggered mechanism, and analyzes closed-loop stability. Section 5 demonstrates the controllers performance, and Section 6 concludes the paper.

Notions: Let $\mathbb{R}$ represent the set of real numbers. $\mathbb{R}^n$ represents an n-dimensional vector space. Let $\mathbb{R}^{m\times n}$ denote the set of all $m\times n$ matrices, with A^T representing the transpose of matrix A. The symbol $\|\cdot\|$ indicates the Euclidean norm for vector and $|\cdot|$ represents absolute value.

3. Preliminaries and problem formulation

3.1. System description

For a nonlinear MAS with N agents, the mth agent dynamics can be expressed as:

$$\begin{aligned} \dot{\mathcal{X}}_{m,k} &= \mathcal{X}_{m,k+1} + \mathcal{F}_{m,k}(\bar{\mathcal{X}}_{m,k}) \\ \dot{\mathcal{X}}_{m,n} &= \mathcal{U}_m + \mathcal{F}_{m,n}(\bar{\mathcal{X}}_{m,n}) \\ \mathcal{Y}_m &= \mathcal{X}_{1,m}, \end{aligned} \tag{1}$$

where $1 \le m \le N$, $1 \le k \le n-1$, $\mathcal{X}_{m,k} \in \mathbb{R}$ and $\mathcal{X}_{m,n} \in \mathbb{R}$ denote the state variables. Additionally, $\bar{\mathcal{X}}_{m,k} = [\mathcal{X}_{m,1}, \dots, \mathcal{X}_{m,k}]^T$ and $\bar{\mathcal{X}}_{m,k} \in \mathbb{R}^k$ stand for vectors formed by the state variables, respectively. Moreover, $\mathcal{U}_m \in \mathbb{R}$

and $\mathcal{Y}_m \in \mathbb{R}$ indicate the control input and output variables. The functions $\mathcal{F}_{m,k}(\bar{\mathcal{X}}_{m,k})$ and $\mathcal{F}_{m,n}(\bar{\mathcal{X}}_{m,n})$, which take values in $\mathbb{R}$, are unknown but exhibit continuous nonlinear behavior. As the leader, the $(N+1)$th agent provides the desired reference signal $Y_d(t) \in \mathbb{R}$.

Assumption 1. (Wen & Chen, 2023): The reference signal $\mathcal{Y}_l(t)$ is continuously differentiable up to order n_m, both the signal and its derivatives are bounded by prescribed positive constants.

3.2. Graph theory

In this paper, the interaction structure among agents is described by an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. The node set is $\mathcal{V} = \{1, \dots, N\}$, and the edge set is $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. The corresponding adjacency matrix is $A = [a_{i,k}] \in \mathbb{R}^{N\times N}$, where $a_{i,k} = 1$ if agent i has access to the information of agent k; otherwise, $a_{i,k} = 0$. The neighbor set of agent m is defined as $\mathcal{N}_m = \{k \mid (m,k) \in \mathcal{E}\}$. Let D be the diagonal in-degree matrix, and define the Laplacian as $L = D - A$ with a dimension of $\mathbb{R}^{N\times N}$. It is further assumed that at least one agent can directly access the leader's information.

Lemma 1. *(Wen & Chen, 2023) For the undirected graph $\mathcal{G}$, its Laplacian L is assumed to be irreducible. Let $\tilde{L} = L + B$, where $B = \mathrm{diag}\{b_1, b_2, \dots, b_N\}$ is a positive definite diagonal matrix. With this modification, $\tilde{L}$ becomes positive definite.*

Assumption 2. In this paper, the undirected graph $\mathcal{G}$ is connected.

3.3. Neural networks framework

Based on the universal approximation capability of neural networks, a two-layer MLPs is employed to represent the unknown functions $\mathcal{F}(\cdot)$ in the system. The first layer weights are fixed after initialization, while the second layer weights are adaptively updated online. For a compact set $\Omega_\gamma \subset \mathbb{R}^\Lambda$, the neural network approximates $\mathcal{F}(\gamma)$ as:

$$\mathcal{F}(\gamma) = W_2^{T*} \cdot \sigma(W_1^T \gamma + b_1) + b_2^* + \varepsilon(\gamma).$$

The proposed two-layer MLPs architecture operates as follows: The input vector $\gamma \in \mathbb{R}^\Lambda$ represents the system states. This vector is processed through a first layer consisting of L neurons, characterized by fixed weights $W_1 \in \mathbb{R}^{\Lambda\times L}$ and biases $b_1 \in \mathbb{R}^L$. The layer applies a Lipschitz-continuous activation function $\sigma(\cdot)$, such as the hyperbolic Tanh or sigmoid function. The second layer employs adaptable weights $W_2 \in \mathbb{R}^L$ that are continuously updated through learning algorithms to minimize the approximation error $\varepsilon(\gamma)$, where $|\varepsilon(\gamma)| \leq \bar{\varepsilon}$ guarantees bounded approximation accuracy. By maintaining the first layer's feature extraction stability, this structure enables adaptive control via the trainable second layer. It effectively combines deterministic feature mapping with online learning capabilities for dynamic system adaptation. The Lipschitz continuity of $\sigma(\cdot)$ ensures predictable behavior for stability analysis and event-triggering threshold design. Fig. 1 illustrates the neural networks architecture.

Assumption 3. The activation function $\sigma(\cdot)$ of the MLPs satisfies the Lipschitz continuity condition, with a known Lipschitz constant $L_\sigma > 0$, which can be expressed as: $\| \sigma(\gamma) - \sigma(Y) \| \leq L_\sigma \| \gamma - Y \|$ for all γ, $Y \in \Omega_\gamma$. The fixed first-layer parameters $W_1 \in \mathbb{R}^{\Lambda\times L}$ and $b_1 \in \mathbb{R}^L$ are bounded such that $\| W_1 \| \leq M$ and $\| b_1 \| \leq M$ for some $M > 0$ ensuring the feature mapping $\sigma(W_1^T \gamma + b_1)$ remains bounded over the compact set Ω_γ. There exists an arbitrarily small positive constant $\bar{\varepsilon}$, satisfying $|\varepsilon(x)| \leq \bar{\varepsilon}$.

Definition 1. The input $\mathcal{U}_m$ for agent m is deemed admissible within the feasible set U_0 upon satisfying: 1) It is initialized to zero with $U_m(0) = 0$; 2) It guarantees that the error $\xi_m(t)$ of the MASs (1) is confined within finite limits for all time; 3) It simultaneously guarantees that the specified performance function constrained.

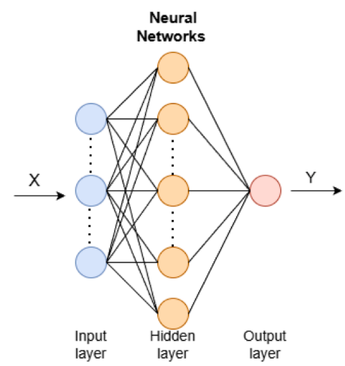

Fig. 1. The architecture of the two-layer multilayer perceptron for adaptive approximation. The structure consists of an input layer for system states, a hidden layer with fixed random weights and biases for stable feature mapping, and an output layer where weights are adaptively updated via the RL mechanism to minimize the approximation error $\epsilon(\gamma)$.

Remark 1. The employed neural networks architecture, with fixed first-layer parameters and adaptively updated second-layer weights, is deliberately chosen to facilitate the subsequent stability analysis and event-triggered control design. By fixing the feature mapping $\sigma(W_1^T \gamma + b_1)$, we ensure that the regression vector remains bounded, which is a crucial prerequisite for deriving the boundedness of adaptive weights W_2. This combined with the Lipschitz continuity of $\sigma(\cdot)$, allows for a tractable design of the event-triggering threshold and the Lyapunov-based stability proof. This approach bypasses the need for the cumbersome RBF center placement in Xu et al. (2023) while retaining the benefits of online learning.

Remark 2. While deep architectures such as convolutional neural networks and Recurrent Neural Networks have achieved success in other domains, MLPs are selected as the function approximators in this work for three specific reasons: 1) The system states are structured physical vectors, not spatial grid data (Li et al., 2024), rendering the convolution operations of convolutional neural networks redundant. 2) The MLPs structure facilitates the formulation of the system into a linearly parameterized form. This allows for the derivation of Lyapunov-based adaptive laws that theoretically guarantee stability, which is significantly more challenging with deep, non-convex architectures. 3) For real-time control of MASs, the inference latency is critical. MLPs offer a streamlined architecture that balances approximation accuracy (guaranteed by the Universal Approximation Theorem) with the low computational cost required for onboard implementation.

Remark 3. In the proposed MLPs-based Actor-Critic framework, the weights of the first layer are randomly initialized and kept fixed during the online adaptation process, while only the output layer weights are updated. This design is motivated by two primary considerations: 1) By fixing the hidden layer, the neural network output becomes linearly parameterized with respect to the adjustable weights W. This allows the use of standard Lyapunov stability theory to derive rigorous adaptive laws, ensuring that the closed-loop signals are semi-globally uniformly ultimately bounded, as detailed in Zhou et al. (2006). 2) For MASs with DETC, fixing the first layer significantly reduces the computational burden per agent. It avoids the "explosion of complexity" typically encountered in backstepping designs when all layers are adaptive, ensuring the real-time feasibility of the DETC-RL scheme. According to the universal approximation theorem, an MLPs with a sufficiently large number of hidden nodes can approximate any continuous nonlinear function to any desired accuracy, even with fixed first-layer parameters, provided they are properly initialized.

Control Objective: For uncertain nonlinear MASs, this paper proposes an optimized distributed control strategy based on both the system states and the errors, which satisfies Assumptions 1–3. Through the integration

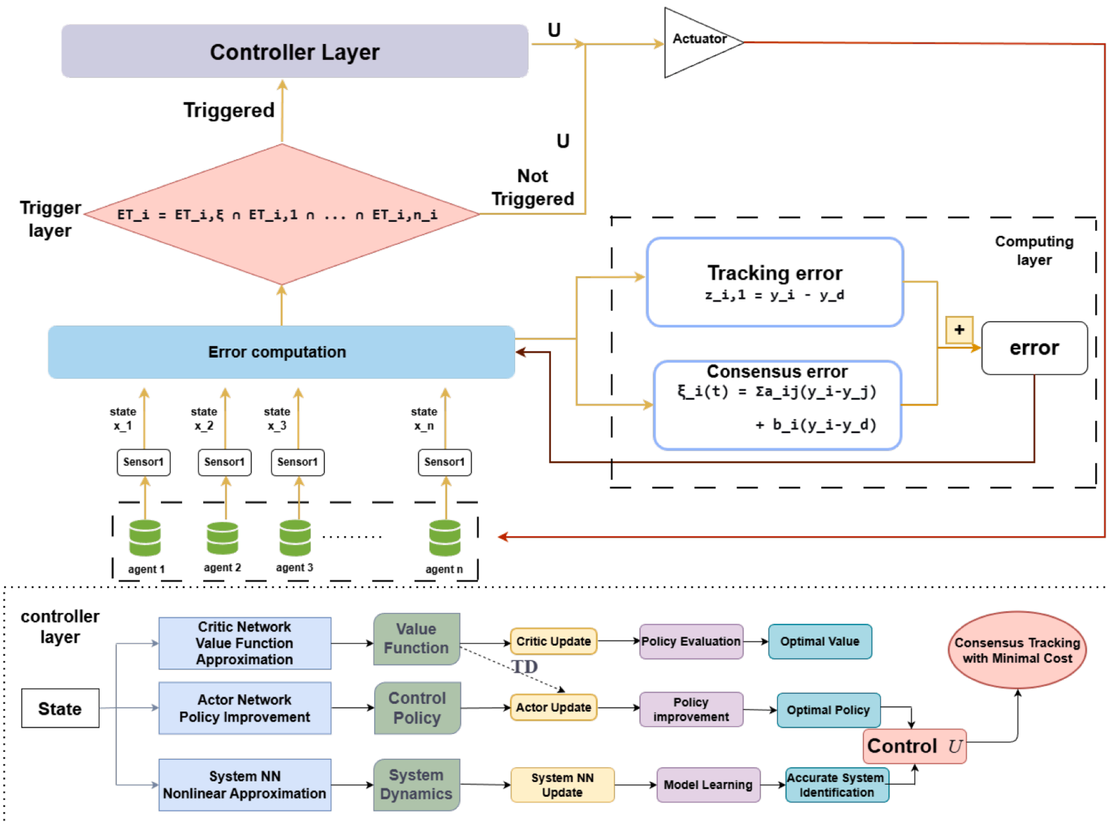

Fig. 2. Complete flowchart. As illustrated in the new figure, the diagram explicitly maps out the information flow: 1.The Critic Network evaluates the current system performance. 2. The Actor Network updates the control weights based on the Critic's feedback. 3. the DETC Module monitors the combined error signal (which contains the current control policy parameterized by the NN weights). The threshold dynamics are thus implicitly adapted online as the neural weights converge, ensuring that the triggering frequency matches the current learning stage.

of RL and DETC, the closed-loop MASs (1) are designed to converge at an exponential rate, with minimal expenditure of control and information exchange.

4. Main results

4.1. Control law development

Fig. 2 presents the overall architecture of the proposed method.

To begin with, the coordinates are transformed as follows:

$$\mathcal{E}_{1,m} = \mathcal{Y}_m - \mathcal{Y}_l, \tag{2}$$

$$\mathcal{E}_{k,m} = \mathcal{X}_{k,m} - \mathcal{U}_{k-1,m},$$

designed in step $(k-1)$, the virtual control law is denoted as $\mathcal{U}_{k-1,m}$.

Step 1: Assume that the unknown yet continuous function $\mathcal{F}_{1,m}(\mathcal{X}_{1,m})$ can be approximated as follows:

$$\mathcal{F}_{1,m}(\mathcal{X}_{1,m}) = W_2^{T*} \cdot \sigma(W_1^T \mathcal{X}_{1,m} + b_1) + b_2^* + \varepsilon_{1,m}(\mathcal{X}_{1,m}),$$

among them, this function can be transformed as:

$$\mathcal{F}_{1,m}(\mathcal{X}_{1,m}) = \theta_{1,m,f}^{T}{}^{*} \varphi_{1,m} + \varepsilon_{1,m}(x_{1,m}),$$

where $\theta_{1,m,f}^{T}{}^{*} = [W_2, b_2]^T \in \mathbb{R}^{L+1}$ denotes the optimal weight vector, $\varphi_{1,m} = [\sigma(W_1^T \mathcal{X}_{1,m} + b_1)^T, 1]^T \in \mathbb{R}^{L+1}$ represents the activation function vector. This error is bounded by $|\varepsilon_{1,m}(X_{1,m})| \le \bar{\varepsilon}_{1,m}$.

The time derivative of $\mathcal{E}_{1,m}$ is:

$$\begin{aligned}\dot{\mathcal{E}}_{1,m} &= \dot{\mathcal{Y}}_m - \dot{\mathcal{Y}}_l \\ &= \mathcal{E}_{2,m} + \mathcal{U}_{1,m} + \theta_{1,m,f}^{T}{}^{*} \varphi_{1,m} + \varepsilon_{1,m} - \dot{\mathcal{Y}}_d.\end{aligned}$$

Furthermore, the definition of the consensus tracking error is

$$\xi_m(t) = b_m(\mathcal{Y}_m(t) - \mathcal{Y}_l(t)) + \sum_{q\in N_m} a_{k,m}(\mathcal{Y}_m(t) - \mathcal{Y}_q(t)).$$

Based on Eq. (2), we have:

$$\xi_m(t) = b_m \mathcal{E}_{1,m}(t) + \sum_{q\in N_m} a_{k,m}(\mathcal{E}_{1,m}(t) - \mathcal{E}_{1,q}(t)).$$

Therefore,

$$\begin{aligned}\dot{\xi}_m(t) =& \mathcal{D}_m(\mathcal{E}_{2,m} + \mathcal{U}_{1,m} + \theta_{1,m,f}^{T}{}^{*} \varphi_{1,m} + \varepsilon_{1,m} - \dot{\mathcal{Y}}_d) - \\ & \sum\nolimits_{q\in N_m} a_{k,m}(\mathcal{E}_{2,m} + \mathcal{U}_{1,m} + \theta_{1,m,f}^{T}{}^{*} \varphi_{1,m} + \varepsilon_{1,m} - \dot{\mathcal{Y}}_d),\end{aligned}$$

where $\mathcal{D}_m = \sum_{q\in N_m} a_{k,m} + b_m$.

The optimal performance function $J_{1,m}^*(\xi_m)$ is defined as follows:

$$\begin{aligned}J_{1,m}^*(\xi_m) &= \int_t^\infty Cost_{1,m}^*(s)ds \\ &= \min_{\mathcal{U}_{1,m}\in\Psi(U_0)} \left(\int_t^\infty Cost(s)ds \right),\end{aligned}$$

here, $Cost_{1,m}^* = \mathcal{U}_{1,m}^{*2} + \xi_m^2$, $\mathcal{U}_{1,m}^*$ denotes the optimal virtual control law. $Cost_{1,m} = \mathcal{U}_{1,m}^2 + \xi_m^2$, where the two terms correspond to the error cost and the virtual control cost, respectively.

To attain optimal control, the corresponding HJB equation is formulated as follows:

$$HJB_{(1,m)} = {\mathcal{U}^{*}_{1,m}}^{2} + \xi_m^2 + \frac{J^{*}_{1,m}(\xi_m)}{d\xi_m}(\mathcal{D}_m(\mathcal{E}_{2,m} + \mathcal{U}_{1,m} + {\theta^{T}_{1,m,f}}^{*}\varphi_{1,m} + \varepsilon_{1,m} - \dot{\mathcal{Y}}_l) - \sum_{q\in N_m} a_{k,m}(\mathcal{E}_{2,m} + \mathcal{U}_{1,m} + {\theta^{T}_{1,m,f}}^{*}\varphi_{1,m} + \varepsilon_{1,m} - \dot{\mathcal{Y}}_l)) = 0.$$

Next, $\mathcal{U}^{*}_{1,m}$ can be derived by solving $\partial HJB_{(1,m)}/\partial\mathcal{U}^{*}_{1,m} = 0$:

$$\mathcal{U}^{*}_{1,m} = -\frac{\mathcal{D}_m}{2}\frac{J^{*}_{1,m}(\xi_m)}{d\xi_m}. \tag{3}$$

It is noteworthy that because the gradient of the optimal performance function $\frac{dJ^{*}_{1,m}(\xi_m)}{d\xi_m}$ is analytically unavailable, the ideal control law (3) is not directly implementable for subsystem optimization. To overcome this challenge, we decompose $\frac{dJ^{*}_{1,m}(\xi_m)}{d\xi_m}$ into the following components:

$$\begin{aligned}\frac{J^{*}_{1,m}(\xi_m)}{d\xi_m} &= \frac{2}{\mathcal{D}_m}(\beta_m\xi_m + {\theta^{T}_{1,m,f}}^{*}\varphi_{1,m} + \varepsilon_{1,m}) + \frac{1}{\mathcal{D}_m}J^{critic}_{1,m}, \\ J^{critic}_{1,m} &= -2(\beta_m\xi_m + {\theta^{T}_{1,m,f}}^{*}\varphi_{1,m} + \varepsilon_{1,m}) + \mathcal{D}_m\frac{J^{*}_{1,m}(\xi_m)}{d\xi_m},\end{aligned} \tag{4}$$

Here, β_m is designed as a positive parameter. Subsequently, the optimal virtual law $\mathcal{U}^{*}_{1,m}$ is reconstructed as:

$$\mathcal{U}^{*}_{1,m} = -\beta_m\xi_m - {\theta^{T}_{1,m,f}}^{*}\varphi_{1,m} - \varepsilon_{1,m} - \frac{1}{2}J^{critic}_{1,m}. \tag{5}$$

Next, since $J^{critic}_{1,m}$ is also unknown, it can likewise be approximated by a neural network.

$$J^{critic}_{1,m} = {\theta^{T}_{1,m,J}}^{*}\psi_{1,m} + \varepsilon_{1,m,J}(\mathcal{X}_{1,m}), \tag{6}$$

here, the error $\varepsilon_{1,m,J}$ satisfies $\bar{\varepsilon}_{1,m,J} \geq |\varepsilon_{1,m,J}(\mathcal{X}_{1,m})|$, and ${\theta^{T}_{1,m,J}}^{*}$ denotes the optimal weight vector.

By substituting (6) into (5) and (4), we obtain:

$$\begin{aligned}\frac{J^{*}_{1,m}(\xi_m)}{d\xi_m} &= \frac{2}{\mathcal{D}_m}(\beta_m\xi_m + {\theta^{T}_{1,m,f}}^{*}\varphi_{1,m} + \varepsilon_{1,m}) + \frac{1}{\mathcal{D}_m}({\theta^{T}_{1,m,J}}^{*}\psi_{1,m} + \varepsilon_{1,m,J}), \\ \mathcal{U}^{*}_{1,m} &= -\beta_m\xi_m - {\theta^{T}_{1,m,f}}^{*}\varphi_{1,m} - \varepsilon_{1,m} - \frac{1}{2}({\theta^{T}_{1,m,c}}^{*}\psi_{1,m} + \varepsilon_{1,m,J}).\end{aligned}$$

To approximate $J^{critic}_{1,m}$, RL techniques are employed.

$$\frac{\hat{J}^{*}_{1,m}(\xi_m)}{d\xi_m} = \frac{2}{\mathcal{D}_m}(\beta_m\xi_m + \hat{\theta}^{T}_{1,m,f}\varphi_{1,m}) + \frac{1}{\mathcal{D}_m}\hat{\theta}^{T}_{1,m,c}\psi_{1,m}, \tag{7}$$

$$\mathcal{U}_{1,m} = -\beta_m\xi_m - \hat{\theta}^{T}_{1,m,f}\varphi_{1,m} - \frac{1}{2}\hat{\theta}^{T}_{1,m,a}\psi_{1,m}, \tag{8}$$

and, $\hat{\theta}_{1,m,c}$ represents the optimal weights for the critic network, and $\hat{\theta}_{1,m,a}$ for the actor network. Furthermore, learning laws are established to update these weight estimations and ensure the convergence of the approximation process:

$$\dot{\hat{\theta}}_{1,m,f} = \xi_m\varphi_{1,m} - \sigma_{1,m,f}\hat{\theta}_{1,m,f}, \tag{9}$$

$$\dot{\hat{\theta}}_{1,m,c} = -\sigma_{1,m,c}\psi_{1,m}\psi^{T}_{1,m}\hat{\theta}_{1,m,c} - \frac{1}{2}\psi_{1,m}\xi_{1,m} + \varkappa_{1,m,c}\sum_{q\in N_m} a_{m,q}(\hat{\theta}_{q,1,c} - \hat{\theta}_{1,m,c}), \tag{10}$$

$$\dot{\hat{\theta}}_{1,m,a} = -\sigma_{1,m,a}\psi_{1,m}\psi^{T}_{1,m}(\hat{\theta}_{1,m,a} - \hat{\theta}_{1,m,c}) + \varkappa_{1,m,a}\sum_{q\in N_m} a_{m,q}(\hat{\theta}_{q,1,a} - \hat{\theta}_{1,m,a}), \tag{11}$$

where $\sigma_{1,m,a}$, $\sigma_{1,m,c}$, $\sigma_{1,m,f}$ and $\varkappa_{1,m,c}$, $\varkappa_{1,m,a}$ are the positive design parameters.

For Layer 1: Lyapunov function is designed using the backstepping technique:

$$\sum_{m=1}^{N} V_{1,m} = \sum_{m=1}^{N}\frac{1}{2}\mathcal{E}_1^T\tilde{L}\mathcal{E}_1 + \sum_{m=1}^{N}\frac{1}{2}\tilde{\theta}^{T}_{1,m,f}\tilde{\theta}_{1,m,f} + \sum_{m=1}^{N}\frac{1}{2}\tilde{\theta}^{T}_{1,m,c}\tilde{\theta}_{1,m,c} + \sum_{m=1}^{N}\frac{1}{2}\tilde{\theta}^{T}_{1,m,a}\tilde{\theta}_{1,m,a},$$

where the estimation errors are defined as $\tilde{\theta}_{1,m,f} = \hat{\theta}_{1,m,f} - \theta^{*}_{1,m,f}$, $\tilde{\theta}_{1,m,c} = \hat{\theta}_{1,m,c} - \theta^{*}_{1,m,J}$, $\tilde{\theta}_{1,m,a} = \hat{\theta}_{1,m,a} - \theta^{*}_{1,m,J}$.

By $V_{1,m}$ and using (8), we obtain the following expression:

$$\begin{aligned}\sum_{m=n}^{N}\dot{V}_{1,m} \leq \sum_{m=1}^{N}\Bigg[&\sum_{l=1,m\neq l}^{N}\xi_m\dot{\mathcal{E}}_{l,1} - \beta_m\xi_m^2 \\ &- \sigma_{1,m}\tilde{\theta}_{1,m,f}\hat{\theta}_{1,m,f} \\ &+ \xi_i(\mathcal{E}_{2,m} - \frac{1}{2}\hat{\theta}^{T}_{1,m,a}\psi_{1,m} + \varepsilon_{1,m} - \dot{\mathcal{Y}}_l) \\ &- \tilde{\theta}^{T}_{1,m,c}\sigma_{1,m,c}\psi_{1,m}\psi^{T}_{1,m}\hat{\theta}_{1,m,c} \\ &- \frac{1}{2}\xi_m\tilde{\theta}^{T}_{1,m,c}\psi_{1,m} \\ &- \tilde{\theta}^{T}_{1,m,a}\sigma_{1,m,a}\psi_{1,m}\psi^{T}_{1,m}(\hat{\theta}_{1,m,a} - \hat{\theta}_{1,m,c})\Bigg] \\ &+ \varkappa_{1,m,c}\sum_{m=1}^{N}\tilde{\theta}^{T}_{1,m,c}\sum_{j\in N_m} a_{ij}(\hat{\theta}_{c,j,1} - \hat{\theta}_{1,m,c}) \\ &+ \varkappa_{1,m,a}\sum_{m=1}^{N}\tilde{\theta}^{T}_{1,m,a}\sum_{j\in N_m} a_{ij}(\hat{\theta}_{a,j,1} - \hat{\theta}_{1,m,a}).\end{aligned}$$

By Young's inequality and $\tilde{\theta}_{1,m,f}$, we obtain:

$$\begin{aligned}\sum_{m=n}^{N}\dot{V}_{1,m} \leq \sum_{m=1}^{N}[&\sum_{l=1,l\neq m}^{N}\xi_m\dot{\mathcal{E}}_{1,l} - (\beta_m - \frac{9}{4})\xi^2 + \frac{\mathcal{E}^2_{2,m}}{2} \\ &- \sigma_{1,m}\tilde{\theta}^{T}_{1,m,f}\tilde{\theta}_{1,m,f} \\ &- (\frac{\sigma_{1,m,c}}{2} - \frac{\sigma_{1,m,a}}{2} - \frac{1}{4})\eta_{1,m}\tilde{\theta}^{T}_{1,m,c}\tilde{\theta}_{1,m,c} \\ &- (\frac{\sigma_{1,m,a}}{2} - \frac{1}{4})\eta_{1,m}\tilde{\theta}^{T}_{1,m,a}\tilde{\theta}_{1,m,a} + D_{1,m}] \\ &- \lambda_2(L)\parallel\tilde{\theta}_c\parallel^2 - \lambda_2(L)\parallel\tilde{\theta}_a\parallel^2 .\end{aligned}$$

where $\eta_{1,m}$ represents the minimum eigenvalue of $\psi_{1,m}\psi^{T}_{1,m}$, $\tilde{\theta}_a$ and $\tilde{\theta}_c$ denote the global vectors of the Actor and Critic networks' weight estimation errors for all agents.

$$D_{1,m} = \frac{\sigma_{1,m,f}}{2}\|\theta^{*}_{1,m,f}\|^2 + \frac{\sigma_{1,m,c}}{2}\lambda_{\max}(\psi_{1,m}\psi^{T}_{1,m})\|\theta^{*}_{1,m,c}\|^2 + \frac{1}{4}\lambda_{\max}(\psi_{1,m}\psi^{T}_{1,m})\|\theta^{*}_{1,m,a}\|^2 + \frac{1}{2}\bar{\varepsilon}^2_{1,m}.$$

Layer ($k = 2, \ldots, n$): The derivative of $\mathcal{E}_{k,m}$ is:

$$\dot{\mathcal{E}}_{k,m} = \mathcal{E}_{k+1,m} + \bar{\mathcal{F}}_{k,m}(v_{k,m}),$$

where $\bar{\mathcal{F}}_{k,m}(v_{k,m}) = \mathcal{F}_{k,m}(\bar{\mathcal{X}}_{k,m}) - \dot{\mathcal{U}}_{k,m-1}$ denotes a continuous and unknown composite function. $v_{k,m} = [\bar{\mathcal{X}}^{T}_{k,m}, \hat{\bar{\rho}}^{T}_{k,m-1}]^T$, and

$$\hat{\bar{\rho}}_{k,m-1} = [\hat{\theta}_{1,m,f}, \ldots, \hat{\theta}_{k,m,f-1}, \hat{\theta}_{1,m,a}, \ldots, \hat{\theta}_{k,m,a-1}]^T.$$

As in the previous step, the $\bar{f}_{k,m}(v_{k,m})$ can be defined as:

$$\bar{f}_{k,m}(v_{k,m}) = \theta^{*T}_{k,m,f}\varphi_{k,m}(v_{k,m}) + \varepsilon_{k,m},$$

where $\theta^*_{k,m,f}$, $\varphi_{k,m}$ and $\varepsilon_{k,m}$ are defined similarly to (6) with $|\ \varepsilon_{k,m}\ | \leq \bar{\varepsilon}_{k,m}$.

Correspondingly, define $J^*_{k,m}(\mathcal{E}_{k,m})$

$$\begin{aligned} J^*_{k,m}(\mathcal{E}_{k,m}) &= \int_t^\infty Cost^*_{k,m}(s)ds \\ &= \min_{\mathcal{U}_{k,m}\in\Psi(U_0)} \left(\int_t^\infty Cost(s)ds\right), \end{aligned}$$

where $Cost_{k,m} = \mathcal{E}^2_{k,m} + \mathcal{U}^2_{k,m}$ serves as the value function, $\mathcal{U}^*_{k,m}$ is the optimal virtual policy while $\mathcal{U}_{k,m}$ denotes the virtual controller intended for approximation purposes.

Consistent with the previous step, to obtain the optimal virtual control law, the HJB equation is given by:

$$\begin{aligned} HJB_{(k,m)} =& \mathcal{E}^2_{k,m} + \mathcal{U}^{*\,2}_{k,m} + \frac{J^*_{k,m}(\mathcal{E}_{k,m})}{d\mathcal{E}_{k,m}}\mathcal{E}_{k+1,m} + \mathcal{U}^*_{k,m} \\ &+ \theta^{*T}_{k,m,f}\varphi_{k,m} + \varepsilon_{k,m}) \\ =& 0. \end{aligned}$$

$\mathcal{U}^*_{k,m}$ is obtained by solving $\partial H_{k,m}/\partial\mathcal{U}^*_{k,m} = 0$:

$$\mathcal{U}^*_{k,m} = -\frac{1}{2}\frac{J^*_{k,m}(\mathcal{E}_{k,m})}{d\mathcal{E}_{k,m}}.$$

Then, the $dJ^*_{k,m}(\mathcal{E}_{k,m})/d\mathcal{E}_{k,m}$ is partitioned as follows:

$$\begin{aligned} \frac{J^*_{k,m}(\mathcal{E}_{k,m})}{d\mathcal{E}_{k,m}} =& 2(\beta_m\mathcal{E}_{k,m} + \theta^{*T}_{k,m,f}\varphi_{1,m} + \varepsilon_{k,m}) \\ &+ J^{critic}_{k,m}(\nu_{k,m}), \end{aligned} \tag{12}$$

$$\begin{aligned} J^{critic}_{k,m} =& -2(\beta_{k,m}\mathcal{E}_{k,m} + \theta^{*T}_{k,m,f}\varphi_{k,m} + \varepsilon_{k,m}) \\ &+ \frac{J^*_{k,m}(\mathcal{E}_{k,m})}{d\mathcal{E}_{k,m}}, \end{aligned}$$

where $\beta_{k,m} > 0$, $\mathcal{U}^*_{k,m}$ is formulated as:

$$\mathcal{U}^*_{k,m} = -\beta_{k,m}\mathcal{E}_{k,m} - \theta^{T\ *}_{k,m,f}\varphi_{k,m} - \varepsilon_{k,m} - \frac{1}{2}J^{critic}_{k,m}(\nu_{k,m}). \tag{13}$$

By using neural network, we can approximate $J^{critic}_{k,m}(\nu_{k,m})$:

$$J^{critic}_{k,m}(\nu_{k,m}) = \theta^{*T}_{k,m,j}\psi_{k,m}(\nu_{k,m}) + \varepsilon_{k,m,j}, \tag{14}$$

the definitions of these parameters are identical to those provided in the preceding step.

By substituting (14) into (12), (13), and referring to (7), (8), the following is obtained,

$$\frac{\hat{J}^*_{k,m}(\mathcal{E}_{k,m})}{dz_{m,q}} = 2(\beta_{k,m}\mathcal{E}_{k,m} + \hat{\theta}^T_{k,m,f}\varphi_{k,m}) + \hat{\theta}^T_{k,m,c}\psi_{k,m},$$

$$\mathcal{U}_{k,m} = -\beta_{k,m}z_{m,q} - \hat{\theta}^T_{k,m,f}\varphi_{k,m} - \frac{1}{2}\hat{\theta}^T_{k,m,a}\psi_{k,m}.$$

The learning rates for the neural network weights are given as follows:

$$\dot{\hat{\theta}}_{k,m,f} = \mathcal{E}_{k,m}\varphi_{k,m} - \sigma_{k,m,f}\hat{\theta}_{k,m,f}, \tag{15}$$

$$\begin{aligned} \dot{\hat{\theta}}_{k,m,c} =& -\sigma_{k,m,c}\psi_{k,m}\psi^T_{k,m}\hat{\theta}_{k,m,c} - \frac{1}{2}\psi_{k,m}\mathcal{E}_{k,m} \\ &+ \varkappa_{k,m,c}\sum_{j\in N_m} a_{m,j}(\hat{\theta}_{c,j,q} - \hat{\theta}_{k,m,c}), \end{aligned} \tag{16}$$

$$\begin{aligned} \dot{\hat{\theta}}_{k,m,a} =& -\sigma_{k,m,a}\psi_{k,m}\psi^T_{k,m}(\hat{\theta}_{k,m,a} - \hat{\theta}_{k,m,c}) \\ &+ \varkappa_{k,m,a}\sum_{j\in N_m} a_{m,j}(\hat{\theta}_{a,j,q} - \hat{\theta}_{k,m,a}), \end{aligned} \tag{17}$$

where $\sigma_{k,m,a}$, $\sigma_{k,m,c}$, $\sigma_{k,m,f}$ and $\varkappa_{k,m,c}$, $\varkappa_{k,m,a}$ are the positive design parameters, and $\mathcal{U}_{m,n_i} = \mathcal{U}_m$, $\mathcal{U}^*_{m,n_i} = \mathcal{U}^*_m$.

Design the Lyapunov function as:

$$\begin{aligned} \sum_{m=1}^N V_{k,m} =& \sum_{m=1}^N V_{k-1,m} + \sum_{m=1}^N \frac{1}{2}\mathcal{E}^2_{k,m} + \sum_{m=1}^N \frac{1}{2}\tilde{\theta}^T_{k,m,f}\tilde{\theta}_{k,m,f} \\ &+ \sum_{m=1}^N \frac{1}{2}\tilde{\theta}^T_{k,m,c}\tilde{\theta}_{k,m,c} + \sum_{m=1}^N \frac{1}{2}\tilde{\theta}^T_{k,m,a}\tilde{\theta}_{k,m,a}, \end{aligned}$$

where $\tilde{\theta}_{k,m,c} = \hat{\theta}_{k,m,c} - \theta^*_{k,m,j}$, $\tilde{\theta}_{k,m,a} = \hat{\theta}_{k,m,a} - \theta^*_{k,m,j}$.

Following the analytical process in the previous step, the following results are derived:

$$\begin{aligned} \sum_{m=n}^N \dot{V}_{1,m} \leq& \sum_{m=1}^N [\dot{V}_{k-1,m} + \frac{\mathcal{E}^2_{k+1,m}}{2} - (\beta_m - \frac{7}{4})\mathcal{E}^2_{k,m} + \frac{\mathcal{E}^2_{2,m}}{2} \\ &- \sigma_{k,m}\tilde{\theta}^T_{k,m,f}\tilde{\theta}_{k,m,f} \\ &- (\frac{\sigma_{k,m,c}}{2} - \frac{\sigma_{k,m,a}}{2} - \frac{1}{4})\eta_{k,m}\tilde{\theta}^T_{k,m,c}\tilde{\theta}_{k,m,c} \\ &- (\frac{\sigma_{k,m,a}}{2} - \frac{1}{4})\eta_{k,m}\tilde{\theta}^T_{k,m,a}\tilde{\theta}_{k,m,a} + D_{k,m}] \\ &- \lambda_2(L)\parallel\tilde{\theta}_c\parallel^2 - \lambda_2(L)\parallel\tilde{\theta}_a\parallel^2, \end{aligned}$$

where η represents the minimum eigenvalue of $\psi_{k,m}\psi^T_{k,m}$, $\tilde{\theta}_a$ and $\tilde{\theta}_c$ denote the global vectors formed by stacking the weight estimation errors of the Actor and Critic neural networks for all agents.

$$\begin{aligned} D_{k,m} =& \frac{\sigma_{k,m,f}}{2}\|\theta^*_{k,m,f}\|^2 + \frac{\sigma_{k,m,c}}{2}\lambda_{\max}(\psi_{k,m}\psi^T_{k,m})\|\theta^*_{k,m,c}\|^2 \\ &+ \frac{1}{4}\lambda_{\max}(\psi_{k,m}\psi^T_{k,m})\|\theta^*_{k,m,a}\|^2 + \frac{1}{2}\bar{\varepsilon}^2_{k,m}. \end{aligned}$$

The RL algorithm is presented below.

Algorithm 1 Actor-critic RL for high-order MASs.

procedure MAIN
  Initialize: Agents $m \in \{1,\dots,N\}$, communication graph $\mathcal{G}$, and feature extractors.
  Define: $K_{\max}$ as the maximum iterations;
  $\theta^m_{f,q}$: identifier weights; $\theta^m_{c,q}$: critic weights; $\theta^m_{a,q}$: actor weights.
  Set hyperparameters $\{\sigma_{f,q}, \sigma_{c,q}, \sigma_{a,q}\}$, κ_c, κ_a
  for $m = 1$ to N **do**
    for $q = 1$ to M **do**
      Initialize $\theta^m_{f,q}, \theta^m_{c,q}, \theta^m_{a,q}$
    end for
  end for
  for $k = 1$ to $K_{\max}$ **do** ▷ Stopping criterion: maximum iterations
    for $m = 1$ to N **do**
      for $q = 1$ to M **do**
        Compute features $\varphi_{q,m}$ and $\psi_{q,m}$ via feat$_q(\mathbf{z}_{q,m})$
        Update $\theta^m_{f,q}$, $\theta^m_{c,q}$, $\theta^m_{a,q}$ via distributed rules (15)–(17)
      end for
    end for
  end for
end procedure

4.2. DETC mechanism

Under the DETC framework, the system states are sampled only at irregular triggering instants, which prevents the controller from relying on continuous state measurements. To address this limitation, the actual and virtual control laws derived from the backstepping design are reformulated to depend solely on the most recently sampled data at the latest triggering instant.

$$\begin{aligned} u_m =& -\beta_{n,m}\check{z}_{n,m} - \hat{\theta}^T_{n,m,f}\varphi_{n,m} \\ &- \frac{1}{2}\hat{\theta}^T_{n,m,a}\psi_{n,m}, \end{aligned} \tag{18}$$

where $\beta_{n,m} > 0$. Moreover, the variables $\check{\mathcal{E}}_{k,m}$, $\check{\mathcal{U}}_{1,m}$, and $\check{\mathcal{U}}_{k,m}$ are:

$$\check{\mathcal{E}}_{k,m} = \check{\mathcal{X}}_{k,m} - \check{\mathcal{U}}_{k-1,m}, \tag{19}$$

$$\check{\mathcal{U}}_{1,m} = -\beta_m \check{\xi}_m - \hat{\theta}_{1,m}^T \varphi_{1,m}(\check{\mathcal{X}}_{1,m}) - \frac{1}{2}\hat{\theta}_{1,m,a}^T \psi_{1,m,j}(\check{\nu}_{1,m}), \tag{20}$$

$$\begin{aligned}\check{\mathcal{U}}_{k,m} =& -\beta_{k,m}\check{\mathcal{E}}_{k,m} - \hat{\theta}_{k,m}^T \varphi_{k,m}(\check{\nu}_{k,m}) \\ &- \frac{1}{2}\hat{\theta}_{k,m,a}^T \psi_{k,m,j}(\check{\nu}_{k,m}), k = 2, \dots, n.\end{aligned} \tag{21}$$

Here, the triggering instants are denoted by $t_{m,l}$ (representing the lth triggering event). The sampled values remain constant within the interval $[t_{m,l}, t_{m,l+1})$ until the next triggering instant occurs. $\check{\mathcal{X}}_{k,m}(t) = \mathcal{X}_{k,m}(t_{m,l})$, and $\check{\xi}_m(t) = \xi_m(t_{m,l})$. Eqs. (18)-(21) respectively present the sampled formulations of the first-layer actual control law, the kth layer virtual control law, and the actual control input.

Subsequently, triggering mechanisms for state signals, which are based on both nodes and edges, are designed:

$$DET_m = DET_{0,m} \cap DET_{1,m} \cap, \dots, \cap DET_{n,m}, \tag{22}$$

where $DET_{0,m}$, $DET_{1,m}$, $DET_{k,m}$ can be expressed as:

$$\begin{cases} DET_{0,m} : DET_{0,m}(t) \geq \mid e_{\xi_m} \mid, \\ DET_{1,m} : DET_{1,m}(t) \geq \mid e_{x_{1,m}} \mid, \\ DET_{k,m} : DET_{k,m}(t) \geq \| \bar{e}_{x_{k,m}} \|, q = 2, \dots, n_m, \end{cases}$$

with:

$$\begin{cases} DET_{0,m}(t) = \frac{B_{m,\xi}}{\beta_m + \frac{1}{2} L_{\sigma_m} \|\hat{\theta}_{1,m,a}(t)\| + \gamma_\xi |\xi_m(t)|} \\ DET_{1,m}(t) = \frac{B_{1,m}}{C}, \\ DET_{k,m}(t) = \frac{B_{k,m}}{C_2}, \end{cases}$$

where, $C = (\beta_m + \frac{1}{2} L_{\sigma_m} \| \hat{\theta}_{1,m,a}(t) \|) u_m + L_{\sigma_m} \| \hat{\theta}_{1,m,f}(t) \| + \gamma_1 \mid \mathcal{E}_{1,m}(t) \mid$. $C_2 = \beta_m + \| \hat{\theta}_{k,m,a}(t) \| + \| \hat{\theta}_{k,m,f}(t) \| + \gamma_k \mid \mathcal{E}_{k,m}(t) \mid$. For $t \in [t_{m,l}, t_{m,l+1})$, we define the error variables as follows: $e_{\xi_m} = \xi_m - \check{\xi}_m$, $e_{x_{k,m}} = \mathcal{X}_{k,m} - \check{\mathcal{X}}_{k,m}$, and $\bar{e}_{x_{k,m}} = \left[e_{x_{1,m}}, \dots, e_{x_{k,m}}\right]^T$. According to the triggering condition (22), once this condition is no longer satisfied, the system will trigger at time $t_{m,l+1}$. The pseudocode for DETC is shown in Algorithm 2.

Algorithm 2 Discrete-time event-triggered mechanism for agent m.

Input: Current states $\mathbf{x}_m(t)$ and $\xi_m(t)$; Last triggered states $\hat{\mathbf{x}}_m$ and $\hat{\xi}_m$.
Parameters: Threshold constants p_ξ, p_1, p_2, p_3; Sampling interval Δt.
loop (at each time step $k\Delta t$)
    Compute measurement errors:
        $e_{\xi,m}(t) \leftarrow \xi_m(t) - \hat{\xi}_m$
        $\mathbf{e}_{x,m}(t) \leftarrow \mathbf{x}_m(t) - \hat{\mathbf{x}}_m$
    Calculate adaptive thresholds E_ξ, E_1, E_2, E_3 via Eq. (22):
        e.g., $E_\xi \leftarrow \frac{p_\xi}{\beta_1 + 0.5 L_\sigma \|\theta_{a,1}^m\| + \gamma_\xi |\xi_m|}$
    Check triggering conditions:
    if $(|e_{\xi,m}| > E_\xi)$ **or** $(|e_{x,m}^{(1)}| > E_1)$ **or** $(\|\mathbf{e}_{x,m}^{(1:2)}\| > E_2)$ **or** … **or** $(\|\mathbf{e}_{x,m}^{(1:k)}\| > E_k)$ **then**
        Trigger: Update $\hat{\mathbf{x}}_m \leftarrow \mathbf{x}_m(t)$ and $\hat{\xi}_m \leftarrow \xi_m(t)$
        Broadcast updated states to neighbors and refresh u_m
    else
        Hold: Maintain current $\hat{\mathbf{x}}_m, \hat{\xi}_m$ and keep u_m constant
    end if
end loop

Remark 4. The primary advantage of incorporating tracking error into the event-triggering threshold, compared to the original weight-dependent mechanism (Xu et al., 2023), lies in its enhanced performance awareness. While the original method conserves resources based solely on the learning state, the modified scheme directly links triggering decisions to control quality. When tracking error increases, the threshold automatically tightens to enforce more frequent control updates, leading to faster error suppression, improved transient performance, and stronger disturbance rejection.

Remark 5. Unlike the low-order systems in Mu et al. (2022) and Liu et al. (2023), the MASs model in this paper is more general. This work investigates high-order systems with uncertain nonlinear dynamics, thereby improving the model's alignment with practical applications. Compared with Wang et al. (2021) and Li et al. (2021), we adopt DETC and design dynamic thresholds by combining update rates. The threshold changes dynamically as the update rate varies, thus reducing resource waste.

Lemma 2. *The magnitudes of the discrepancies caused by the triggering scheme,* $\mid \mathcal{U}_{k,m} - \check{\mathcal{U}}_{k,m} \mid$ *and* $\mid \mathcal{U}_i - \mathcal{U}_{n,m} \mid$ *are upper-bounded as:*

$$\mid \mathcal{U}_{k,m} - \check{\mathcal{U}}_{k,m} \mid \leq \bar{B}_{k,m}, \mid \mathcal{U}_i - \mathcal{U}_{n,m} \mid \leq \bar{B}_{n,m},$$

here $\bar{B}_{k,m} > 0$, $\bar{B}_{n,m} > 0$ *represent constant bounds.*

Proof. Based on the dynamic event-triggering mechanism and Assumption 3, the following relation is derived:

$$\begin{aligned}\mid \mathcal{U}_{1,m} - \check{\mathcal{U}}_{1,m} \mid \leq& \mid e_{\xi_m} \mid (\beta_m + \frac{1}{2} L_{\sigma_m} \| \hat{\theta}_{1,m,a} \|) + \mid e_{x_i,1} \mid \\ & \times (L_{\sigma_m} \| \hat{\theta}_{1,m,f} \| + \frac{1}{2} L_{\sigma_m} \| \hat{\theta}_{1,m,a} \|) \\ \leq& \bar{B}_{1,m},\end{aligned}$$

where $L_{\sigma_m} = \max\{L_{\varphi_{1,m}}, \dots, L_{\varphi_{m,n}}, L_{\psi_{j,1,m}}, \dots, L_{\psi_{j,m,n}}\}$, $\bar{B}_{1,m} = B_{\xi_m} + B_{1,m} > 0$. Subsequently, this gives

$$\mid \mathcal{E}_{2,m} - \check{\mathcal{E}}_{2,m} \mid \leq \mid e_{x_{2,m}} \mid + \bar{B}_{1,m}.$$

The procedure for layer k involves the repeated application of the above process.

$$\begin{aligned}\mid \mathcal{U}_{k,m} - \check{\mathcal{U}}_{k,m} \mid \leq& \beta_{k,m}(\| \bar{e}_{x_{k,m}} \| + \bar{p}_{k,m-1}) + L_{\sigma_m} \| \hat{\theta}_{k,m,f} \| \\ & \times \| \bar{e}_{x_{k,m}} \| + \frac{1}{2} L_{\sigma_m} \| \hat{\theta}_{k,m,a} \| \| \bar{e}_{x_m,k} \| \\ \leq& \bar{B}_{k,m},\end{aligned}$$

here, $\bar{B}_{k,m} = \sum_{k=1}^{k-1} \beta_{k+1,m} \bar{B}_{k,m} + B_{k,m}$ is a constant.

Subsequently, it can be derived that:

$$\mid \mathcal{E}_{k+1,m} - \check{\mathcal{E}}_{k+1,m} \mid \leq \mid e_{x_{k+1,m}} \mid + \bar{B}_{k,m}.$$

For the highest layer n_m, the boundedness of the actual control input relative to the optimal virtual control is expressed as:

$$\begin{aligned}\mid \mathcal{U}_m - \mathcal{U}_{n,m} \mid \leq& \beta_{n,m}(\| \bar{e}_{x_{n,m}} \| + \bar{B}_{n,m}) + L_{\sigma_m} \| \hat{\theta}_{n,m,f} \| \\ & \times \| \bar{e}_{x_{n,m}} \| + \frac{1}{2} L_{\sigma_m} \| \hat{\theta}_{n,m,a} \| \| \bar{x}_{n,x_m} \| \\ \leq& \bar{B}_{n,m},\end{aligned}$$

where $\bar{B}_{n,m} = B_{n,m} + \sum_{k=1}^{k-1} \beta_{k+1,m} \bar{B}_{n,m}$, and $\bar{B}_{n,m}$ is a constant. □

Remark 6. It is worth noting the robustness of the proposed triggering mechanism during the transient learning phase. Since the dynamic triggering threshold is continuously coupled with the learned Actor-Critic neural weights, one might be concerned that transient weight oscillations could induce erratic triggering behavior or even Zeno phenomena. However, the proposed framework is inherently robust against such oscillations. The Lyapunov-derived weight tuning laws utilize a robust σ term, which theoretically guarantees that all weight estimates remain strictly bounded within a compact set (SGUUB). Consequently, the adaptive threshold is prevented from uncontrolled expansion or collapsing to zero. This mathematically bounded nature ensures that the derived Minimum Inter-Event Time remains strictly positive, effectively preventing pathological triggering bursts and excluding Zeno behavior even during the initial active learning phase.

4.3. Stability analysis

Theorem 1. *Consider the uncertain nonlinear MASs (1) over a connected and undirected interaction graph. Suppose that Assumptions 1 and 2 hold, and the following conditions are satisfied:*

(i) The control law is implemented within a RL mechanism employing an Actor-Critic architecture, with update rules given by (10), (16), (11), and (17).

(ii) Neural networks with update laws (9) and (15) are utilized to approximate the completely unknown nonlinearities.

(iii) The virtual and actual controllers are designed according to (20), (21), and (18), under the event-triggered mechanism (22).

Then, the resulting closed-loop system guarantees that:

(i) All internal signals are semi-globally uniformly ultimately bounded.

(ii) The deviation between the dynamic event-triggered controller and its optimal counterpart remains within a tunable bound.

Proof. To analyze the stability of the overall closed-loop system, we construct the following global candidate Lyapunov function:

$$V = \sum_{m=1}^{N} V_{k,m},$$

where $V_{k,m}$ represents the Lyapunov function component for the kth layer of agent m.

$$\sum_{m=1}^{N} V_{k,m} = \sum_{m=1}^{N} V_{k-1,m} + \sum_{m=1}^{N} \frac{1}{2}\mathcal{E}_{k,m}^2 + \sum_{m=1}^{N} \frac{1}{2}\tilde{\theta}_{n,m,f}^T\tilde{\theta}_{n,m,f} + \sum_{m=1}^{N} \frac{1}{2}\tilde{\theta}_{n,m,c}^T\tilde{\theta}_{n,m,c} + \sum_{m=1}^{N} \frac{1}{2}\tilde{\theta}_{n,m,a}^T\tilde{\theta}_{n,m,a}.$$

Then,

$$\begin{aligned}\sum_{m=1}^{N} \dot{V}_{k,m} = \sum_{m=1}^{N} \dot{V}_{k,m-1} &+ \sum_{m=1}^{N}[\mathcal{E}_{k,m}(\mathcal{U}_m - \mathcal{U}_{k,m} - \beta_{k,m}\mathcal{E}_{k,m} - \frac{1}{2}\hat{\theta}_{k,m,a}^T\varphi_{k,m} + \varepsilon_{k,m}) - \sigma_{k,m,f}\tilde{\theta}_{k,m,f}\hat{\theta}_{k,m,f}\\
&- \tilde{\theta}_{k,m,c}^T\sigma_{k,m,c}\psi_{k,m}\psi_{k,m}^T\hat{\theta}_{k,m,c} - \frac{1}{2}\tilde{\theta}_{k,m,c}^T\psi_{k,m}\mathcal{E}_{k,m} - \tilde{\theta}_{k,m,a}^T\sigma_{k,m,a}\psi_{k,m}\psi_{k,m}^T(\hat{\theta}_{k,m,a} - \hat{\theta}_{k,m,c})]\\
&+ \varkappa_{k,m,c}\sum_{m=1}^{N}\tilde{\theta}_{k,m,c}^T\sum_{q\in k} a_{ij}(\hat{\theta}_{k,q,c} - \hat{\theta}_{k,m,c}) + \varkappa_{k,m,a}\sum_{m=1}^{N}\tilde{\theta}_{k,m,a}^T\sum_{q\in k} a_{ij}(\hat{\theta}_{k,q,a} - \hat{\theta}_{k,m,a}).\end{aligned}$$

Using Young's inequality results in:

$$\begin{aligned}\sum_{m=1}^{N} \dot{V}_{n,m} \le \sum_{m=1}^{N}\Bigg[&\dot{V}_{n-1,m} + \frac{\bar{B}_{n,m}^2}{2} - \left(\beta_{n,m} - \frac{7}{4}\right)\mathcal{E}_{n,m}^2 - \frac{\sigma_{n,m,f}}{2}\|\tilde{\theta}_{n,m,f}\|^2\\
&- \left(\frac{\sigma_{n,m,c}}{2} - \frac{\sigma_{n,m,a}}{2} - \frac{1}{4}\right)\eta_{n,m}\|\tilde{\theta}_{n,m,c}\|^2 - \left(\frac{\sigma_{n,m,a}}{2} - \frac{1}{4}\right)\eta_{n,m}\|\tilde{\theta}_{n,m,a}\|^2 + D_{n,m}\Bigg]\\
&- \underline{\lambda}(L)\|\tilde{\theta}_c\|^2 - \underline{\lambda}(L)\|\tilde{\theta}_a\|^2.\end{aligned}$$

Lastly,

$$\sum_{m=1}^{N} \dot{V}_{k,m} \le \sum_{m=1}^{N}\Big(\sum_{l=1,l\neq m}^{N} \xi_m\dot{\mathcal{E}}_{l,1} - (\beta_m - \frac{9}{4})\xi_m^2\Big) - \sum_{m=1}^{N} C_m V_{n,m} + \sum_{m=1}^{N}\Big(\sum_{k=1}^{n} D_{k,m} + \frac{\bar{B}_{k,m}^2}{2}\Big), \tag{23}$$

with

$$\begin{aligned}C_m = min\{&2(\beta_{1,m} - \frac{7}{4}), \dots, 2(\beta_{n,m} - \frac{7}{4}), 2\sigma_1, \dots, 2\sigma_n,\\
&2(\frac{\sigma_{1,m,c}}{2} - \frac{\sigma_{1,m,a}}{2} - \frac{1}{4}), \dots, 2(\frac{\sigma_{n,m,c}}{2} - \frac{\sigma_{n,m,a}}{2} - \frac{1}{4}),\\
&2(\frac{\sigma_{1,m,a}}{2} - \frac{1}{4}), \dots, (\frac{\sigma_{n,m,a}}{2} - \frac{1}{4}), \lambda_2(L)\},\end{aligned}$$

Using Young's inequality results in:

$$\sum_{l\in\mathcal{N}_m} \xi_m\dot{\mathcal{E}}_{l,1} \le \frac{1}{4}\xi_m^2 + \left(\sum_{l\in\mathcal{N}_m} \dot{\mathcal{E}}_{l,1}\right)^2. \tag{24}$$

Substituting (24) into (23), we can obtain:

$$\dot{V} \le -\sum_{m=1}^{N}\left(\beta_m - \frac{5}{2}\right)\xi_m^2 - \sum_{m=1}^{N} C_m V_{n,m} + \sum_{m=1}^{N} D_m,$$

$$D_m = \sum_{k=1}^{n} D_{k,m} + \frac{\bar{B}_{n,m}^2}{2} + \left(\sum_{l\in\mathcal{N}_m} \dot{\mathcal{E}}_{l,1}\right)^2.$$

$$\dot{V} \le -\bar{C}V + D, \tag{25}$$

where $\bar{C} = min\{C_1, \dots, C_N\}$, and $D = \sum_{m=1}^{N} D_m$. This inequality holds when the design parameters satisfy $\beta_m > \frac{5}{2}$, $\beta_{k,m} > \frac{7}{4}$, $\sigma_{k,m,a} > \frac{1}{4}$, $\sigma_{k,m,c} > \sigma_{k,m,a} + \frac{1}{4}$, for $q = 1, \dots, n$.

By integrating inequality (25) over the interval $[0, t]$, we obtain:

$$0 \le V(t) \le \frac{D}{\bar{C}} + [v(0) - \frac{D}{\bar{C}}]e^{-Ct}.$$

This confirms that all signals in the closed-loop system exhibit semi-globally uniformly ultimately bounded stability with a configurable bound. Consequently, the consensus tracking error asymptotically converges to and remains within a compact set defined by $\lim_{t\to\infty}|\xi_m| \le \sqrt{D/\bar{C}}$.

The error between $\mathcal{U}_m$ and $\mathcal{U}_m^*$ is defined as:

$$\begin{aligned}|\mathcal{U}_m - \mathcal{U}_m^*| \le\ &\|e_{x_{m,n}}\|\left(\beta_{n,m} + L_{\sigma_m}\left(\|\hat{\theta}_{n,m,f}\| + \frac{1}{2}\|\hat{\theta}_{n,m,a}\|\right)\right)\\
&+ \|\tilde{\theta}_{n,m,f}\|\|\varphi_{i,n_i}\| + \frac{1}{2}\|\tilde{\theta}_{n,m,a}\| \cdot \|\psi_{i,n_i}\|\\
&+ \beta_{n,m}\tilde{B}_{k,m} + \varepsilon_{n,m,f} + \frac{1}{2}\varepsilon_{n,m,j},\end{aligned}$$

based on the preceding derivation, all the aforementioned parameters are bounded. Therefore, the final result is also bounded. □

Theorem 2. *Under the triggering condition (22), there exists a uniform positive lower bound on the interval between any two consecutive triggering instants of agent m.*

Proof. From Theorem 1, it is known that all signals of the closed-loop system are bounded. In particular,

$$|\dot{\xi}_m(t)| \le \mu_{m,\xi}, \|\dot{\mathcal{X}}_{k,m}(t)\| \le \mu_{k,m},$$

where there exist positive constants $\mu_{m,\xi}$ and $\mu_{k,m}$ as upper bounds. Let

$$\bar{\mu}_m = max\{\mu_{m,\xi}, \mu_{1,m}, \dots, \mu_{k,m}\}.$$

Within the triggering window $[t_{m,l}, t_{m,l+1})$, the error grows over time, and its upper bound can be expressed by an integral:

$$|e_{m,\xi}(t)| \le \int_{t_{m,l}}^{t_{m,l+1}} |\dot{e}_{m,\xi}(s)| ds \le \bar{\mu}_m(t_{m,l+1} - t_{m,l}).$$

The triggering condition requires that the error reaches the threshold $DET_{m,\xi}, \ldots DET_{n,m}$. Let the minimum threshold be defined as:

$$\pi_m := min\{DET_{m,\xi}, DET_{1,m}, \ldots, DET_{n,m}\} > 0.$$

From the error upper bound and the minimum threshold, we have:

$$\bar{\mu}_m(t_{m,l+1} - t_{m,l}) \ge \pi_m,$$

$$t_{m,l+1} - t_{m,l} \ge \frac{\pi_m}{\bar{\mu}} > 0.$$

Therefore, there exists a positive lower bound $\frac{\pi_m}{\bar{\mu}}$ on the inter-event intervals for each agent, which excludes Zeno behavior. □

Remark 7. It is worth noting that the proposed dynamic event-triggered mechanism and the reinforcement learning framework currently operate primarily in the time domain, relying on real-time local state tracking errors. However, complex multi-agent systems often exhibit profound deep global correlations. Exploring frequency-domain decompositions presents a highly promising perspective to capture these complex dynamics. As demonstrated in recent studies (Ouyang et al., 2025), employing spectral graph neural networks for high-frequency and low-frequency filtering can effectively balance global topological information and local state variations. While incorporating such frequency-domain insights to restructure the current Lyapunov-based time-domain framework falls into our future research, it offers a significant avenue to further enhance the optimization performance and learning efficiency of the Actor-Critic networks.

5. Numerical example

5.1. Simulation experiment

To verify the effectiveness of the DETC mechanism proposed in this paper, the electromechanical system from (Xu et al., 2023) is adopted as the research object in the experiment. This system consists of 1 leader agent and 6 follower agents, and the performance of the strategy proposed in this paper is compared with the optimized event-triggered control strategy proposed in Xu et al. (2023). The experimental results show that the framework proposed in this paper has smaller total cost, fewer triggering times, and smaller tracking error than the strategy in Xu et al. (2023) in terms of these three indicators.

$$D\ddot{p} + B\dot{p} + N\sin(p) = \tau_r + \tau_d,$$
$$C\dot{\tau}_r + H\tau_r = u - K_m\dot{p},$$

where p, $\dot{p}$, and $\ddot{p}$ represent the motor angular position, velocity, and acceleration τ_r is the torque provided by the electrical components,and u is the motor torque. For consistency with the parameter settings of the original system, the detailed definitions are provided in Meng et al. (2022). Fig. 3 illustrates the practical model of the electromechanical system.

Consistent with Xu et al. (2023), the system state variables are defined as follows: $r_1 = CDp$, $r_2 = CD\dot{p}$, $r_3 = C\tau_r$. The dynamic equation of the mth $(m = 1, \ldots, 6)$ agent can be expressed as follows:

$$y_m = r_{1,m},$$
$$\dot{r}_{3,m} = u_i + \mathcal{F}_{3,m}(\bar{r}_{3,m}),$$
$$\dot{r}_{2,m} = r_{3,m} + \mathcal{F}_{2,m}(\bar{r}_{2,m}),$$
$$\dot{r}_{1,m} = r_{2,m} + \mathcal{F}_{1,m}(\bar{r}_{1,m}).$$

the specific relevant system parameters herein are consistent with those in reference [2]. Finally, the leader's signal is set to $y = \sin(2t)$.

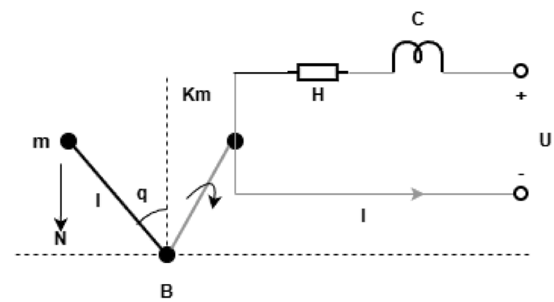

Fig. 3. Model of the practical electromechanical system.

To enable data comparison with [2], the initial parameters in this paper are set to be the same as those in [2]. Specifically, the initial values for each agent are $r_{1,m}(0) = [0.5, 1.2, -0.2, -0.6, 1.0, -0.9]^T$, $r_{2,m}(0) = [0, 0, 0, 0, 0, 0]^T$, $r_{3,m}(0) = [0, 0, 0, 0, 0, 0]$. For the two-layer MLPs neural network framework, the weights of the first layer remain fixed. W_1 is sampled from a normal distribution with a mean of 0.8 and a standard deviation of 0.4, while the bias term b_1 is sampled from a normal distribution with a mean of 0 and a standard deviation of 0.2. The tanh function is selected as the activation function, and w_2 and b_2 are obtained through training. The simulation is carried out for 20 s with a time increment of 0.002 s. The remaining parameter values are: $\sigma_{1,m,c} = 50$, $\sigma_{2,m,c} = 50$, $\sigma_{3,m,c} = 50$, $\sigma_{1,m,a} = 25$, $\sigma_{2,m,a} = 30$, $\sigma_{3,m,a} = 40$, $\sigma_{1,m,f} = 10$, $\sigma_{2,m,f} = 12$, $\sigma_{3,m,f} = 40$, $\kappa_{c,m,n} = 0.3$, $\kappa_{a,m,n} = 0.3$, $\beta_{1,m} = 14$, $\beta_{2,m} = 30$, $\beta_{3,m} = 70$, $B_{m,\xi} = 1.2$, $B_{1,m} = 0.6$, $B_{2,m} = 1.2$, $B_{m,3} = 1.3$. Although the proposed scheme involves several design parameters, their selection follows a systematic procedure based on their physical significance: 1) The control gains κ_c and κ_a should be tuned first. Larger gains improve tracking speed but may induce high-frequency control actions. They are typically increased until a balance between convergence rate and control smoothness is reached. 2) The triggering thresholds β and γ regulate the event-triggering frequency. β determines the sensitivity to state deviations. A practical approach is to start with a small β (dense communication) and gradually increase it to reduce bandwidth usage until the maximum acceptable tracking error is reached. 3) The learning rate σ governs the speed of neural weight adaptation. It should be set sufficiently high to capture dynamics but low enough to avoid oscillation.

Figs. 4–12 present the simulation results for the electromechanical system. Fig. 4 shows that under the control law designed in this paper, each agent exhibits excellent performance in tracking the leader's signal with its output signal. Figs. 5 and 6 respectively displays the variation patterns of velocity and angular acceleration, as well as the output of the control signal u. Figs. 7–9 show the curves of $\theta_{2,m,f}$, $\theta_{3,m,f}$, $\theta_{k,m,c}$ and $\theta_{k,m,a}$. Fig. 10 shows the triggering instants of each agent. Fig. 11 presents a comparison of the control costs incurred by the method proposed in this paper and that in Xu et al. (2023). It shows that the total cost of our method is lower, with the cost of each agent reduced by 20%. Fig. 12 show that the number of triggering events of the event-triggered control designed in this paper is fewer than that of the previous methods, each agent achieves fewer triggers, and the required communication cost is lower.

5.2. Comprehensive performance analysis: Scalability, architecture, and robustness

To comprehensively evaluate the overall efficacy of the proposed control framework, this section conducts an in-depth comparative analysis across *three experiments*: the scalability and consensus benefits under varying network sizes, the superiority of the neural network architecture, and robustness against measurement noise.

First, *experiment 1* simultaneously investigates the system scalability and the communication efficiency of the proposed consensus mecha-

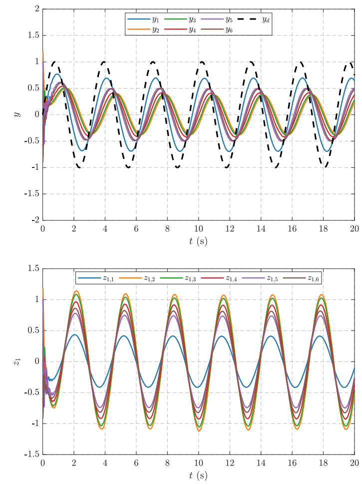

Fig. 4. Outputs y_m and Tracking errors.

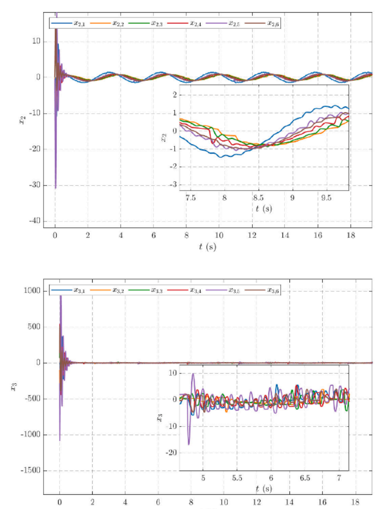

Fig. 5. the variation patterns of velocity and angular acceleration.

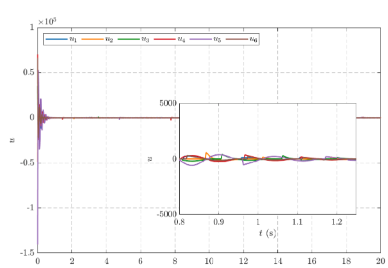

Fig. 6. the variation patterns of output of the control signal u.

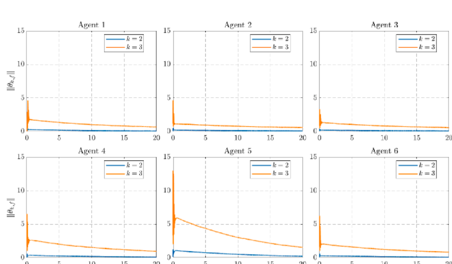

Fig. 7. Norms of $\hat{\theta}_{2,m,f}$ and $\hat{\theta}_{3,m,f}$ update laws with $m = 1, \ldots, 6$.

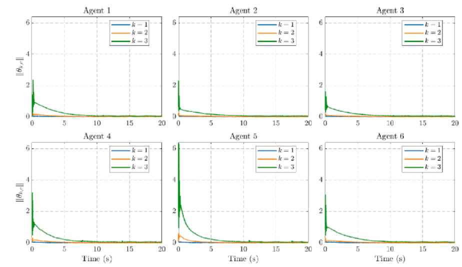

Fig. 8. Norms of $\hat{\theta}_{1,m,c}$, $\hat{\theta}_{2,m,c}$ and $\hat{\theta}_{3,m,c}$ update laws with $m = 1, \ldots, 6$.

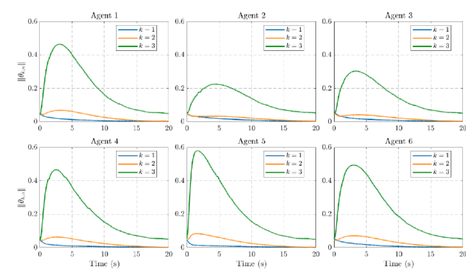

Fig. 9. Norms of $\hat{\theta}_{1,m,a}$, $\hat{\theta}_{2,m,a}$ and $\hat{\theta}_{3,m,a}$ update laws with $m = 1, \ldots, 6$.

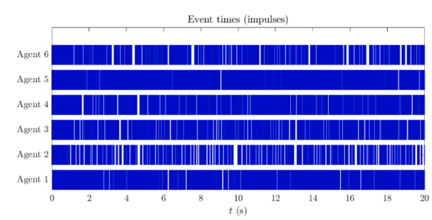

Fig. 10. the triggering instants of each agent.

nism. The simulation was conducted across three network scales: small ($N = 6$), medium ($N = 12$), and large ($N = 20$), comparing the proposed consensus-based DETC-RL method against a non-consensus baseline. As illustrated in Fig. 13 the results reveal a critical trade-off advantage: 1) As shown in the RRMSE comparison, both methods achieve comparable high-precision tracking, and the accuracy is insensitive to the increase in network size ($N = 6$ to $N = 20$). 2) However, the distinct advantage of the proposed framework is highlighted in the triggering count comparison. Despite achieving similar tracking accuracy, the proposed method significantly reduces the cumulative number of triggering events compared to the baseline. This indicates that the proposed consensus-based protocol achieves a superior cost-performance ratio: it maintains high-fidelity synchronization while consuming fewer communication resources, thereby effectively mitigating network congestion in large-scale systems.

Second, to validate the rationale for the neural network architecture selection, we compared the control performance of the proposed MLPs architecture against a traditional RBF network (Zhu et al., 2025) under identical conditions in *experiment 2*. As shown in Fig. 14, due to the susceptibility of RBF networks to the "curse of dimensionality" in high-order systems, they exhibit a relatively large tracking error (RRMSE = 0.8533). In contrast, the proposed MLPs architecture, leveraging its global approximation capability, reduces the RRMSE to 0.5762 (a performance improvement of approximately 32.5%), achieving superior control precision with reduced parameter redundancy.

Finally, considering disturbances in practical environments, Gaussian white noise with a standard deviation of 0.005 was introduced into the output measurements to test system robustness. *Experiment 3* results (Fig. 15) demonstrate that even under noise corruption, the agents successfully achieve precise tracking of the leader. Furthermore, the con-

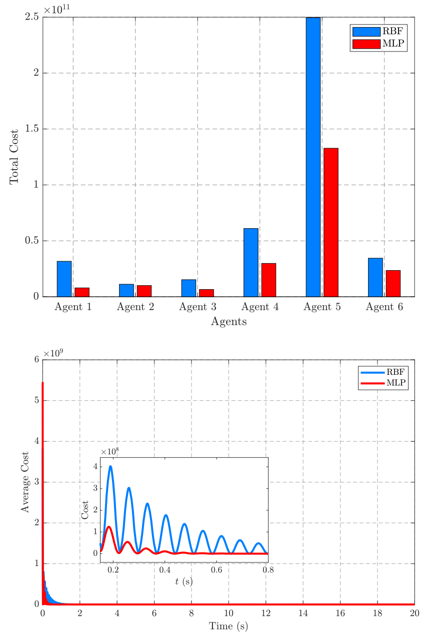

Fig. 11. Average cost comparison and The comparison of the total cost among individual agents.

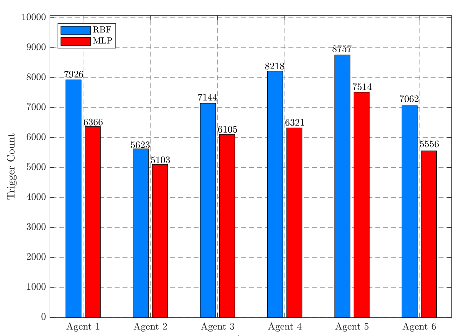

Fig. 12. Comparison of triggering counts.

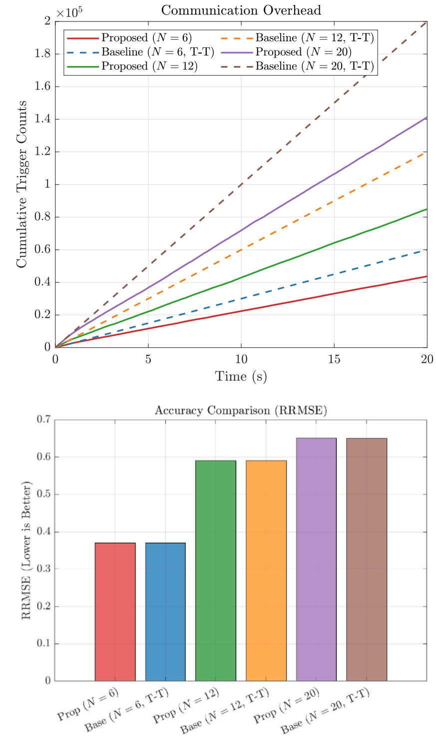

Fig. 13. RRSME and trigger counts for different numbers of agents under consensus and non-consensus conditions in Experiment 1.

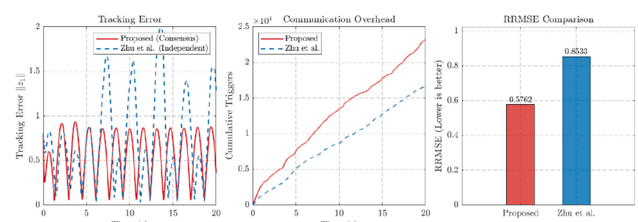

Fig. 14. RBF and MLPs scenarios in Experiment 2.

trol inputs u_i converge rapidly without exhibiting noise-induced high-frequency chattering. This confirms that the σ-modification term in the adaptive law effectively suppresses parameter drift, ensuring the strong robustness of the closed-loop system in non-ideal environments. Furthermore, while the above analysis confirms the system's robustness against sensor noise, it is practically significant to briefly discuss the framework's resilience to network imperfections, such as communication delays or packet loss. In practical scenarios, if a triggered data packet is lost, the neighboring agents will not receive the updated state. Consequently, the local consensus tracking error will continue to accumulate. However, due to the deeply coupled nature of the proposed DETC mechanism, once this accumulating error violates the dynamic triggering condition again, a subsequent triggering event will be immediately generated. Thus, although severe packet loss would naturally increase the overall triggering frequency to compensate for the lost information, the closed-loop stability and the ultimate tracking performance of the multi-agent system are theoretically preserved.

Table 1
Comparison of computational complexity and resource requirements per control step.

Metric	Deep RL	Standard RBF Network	Proposed DETC-RL
Architecture	Deep Multilayer (Fully Adaptive)	Shallow (Gaussian Kernels)	**Single-Hidden Layer (Fixed Base)**
Tunable Parameters	$O(D \cdot L^2)$	$O(m \cdot \mathcal{N}^n)$	$\mathbf{O(m \cdot L)}$
Time Complexity	$O(D \cdot L^2 +$ Backprop)	$O(m \cdot \mathcal{N}^n)$	$\mathbf{O(L \cdot (n + m))}$
Update Mechanism	Gradient Descent (Chain Rule)	Linear Algebra	**Lyapunov-based**
Memory Requirement	High (Replay Buffer + Batches)	Moderate	**Low (Current State Only)**

Note: n: dimension of state vector; m: dimension of control input; L: number of hidden neurons (nodes); D: depth of deep networks; $\mathcal{N}$: number of grids per dimension for RBF.
The proposed method achieves linear scalability $O(n)$ regarding state dimension, whereas RBF suffers from exponential growth $O(\mathcal{N}^n)$.

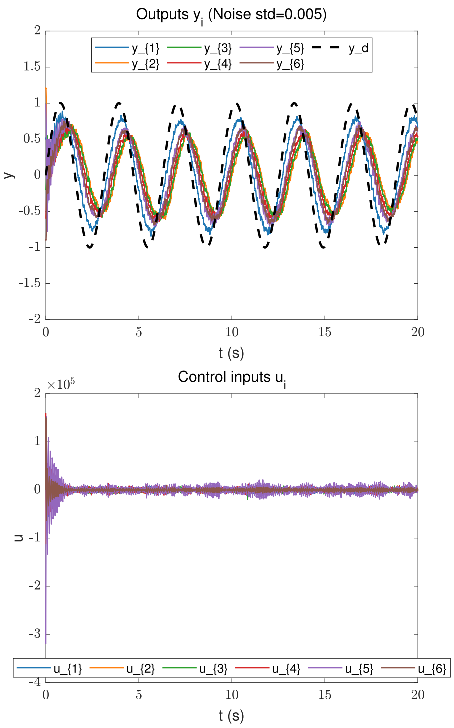

Fig. 15. Experiment diagram with added noise in Experiment 3.

5.3. Complexity and resource overhead analysis

To rigorously assess the computational feasibility of the proposed algorithm for real-time embedded applications, we conducted a theoretical complexity analysis. Table 1 presents a comprehensive comparison of the proposed DETC-RL framework against mainstream Deep RL and standard RBF neural network control schemes. The evaluation metrics include network architecture, the number of tunable parameters, time complexity per control step, and memory requirements.

As highlighted in Table 1, the proposed DETC-RL framework demonstrates superior computational efficiency by addressing the limitations of existing methods. Unlike standard RBF networks that suffer from the "curse of dimensionality" (exponential parameter growth) and Deep RL algorithms that demand intensive backpropagation and large memory buffers, the proposed method achieves linear scalability ($O(L \cdot (n+m))$) and a minimal memory footprint through a deterministic Lyapunov-based update law. This low-complexity design significantly reduces the online learning overhead, ensuring the algorithm is strictly feasible for real-time deployment on resource-constrained embedded platforms.

6. Conclusion

This paper has developed a robust optimized consensus tracking framework for high-order nonlinear MASs by integrating Actor-Critic RL with a dynamic event-triggered mechanism. A key innovation of this study has been the effective reconciliation of high-precision control and communication efficiency. By embedding consensus errors into the RL and utilizing adaptive thresholds, the proposed scheme has significantly reduced communication overhead while ensuring closed-loop stability and excluding Zeno behavior. Simulation results on a multi-electromechanical system have confirmed that the method maintains superior performance and robustness against sensor noise, even under significant communication constraints. Consequently, this work provides a reliable theoretical and practical solution for resource-constrained networked control systems.

Despite these contributions, certain limitations remain. The current framework relies on full state feedback under a fixed undirected topology, and the online training of MLPs involves increased computational complexity. Future research will address these challenges by: 1) Developing lightweight learning algorithms to reduce computational burdens for edge deployment; 2) Extending the framework to handle output feedback control for heterogeneous agents under directed or switching topologies; and 3) Enhancing the system's resilience against intelligent cyber-attacks.

CRediT authorship contribution statement

Xiaoli Ruan: Writing – original draft, Software, Methodology, Conceptualization; **Shaowei Liang:** Writing – original draft, Software, Methodology, Conceptualization; **Tao Peng:** Writing – review & editing, Methodology, Funding acquisition; **Ailong Wu:** Writing – review & editing, Methodology, Funding acquisition; **Ze Tang:** Writing – review & editing, Methodology, Funding acquisition; **Jianwen Feng:** Writing – review & editing, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that there are no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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